

The Districting Integrity Act

Huntington-Hill as the Constitutional Standard for
Intrastate Congressional Apportionment

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Abstract

Every ten years, Congress apportions 435 seats among the fifty states using the Huntington-Hill priority rule—an algorithm whose constitutional pedigree is untested. Yet when each state then divides its seats into districts, no federal standard governs the method. The result is partisan gerrymandering, litigation that consumes a decade, and maps that fail to represent anyone’s constituents well.

This Article proposes a fix grounded entirely in first principles that both parties have already endorsed. The argument is four steps. *First* (*Wesberry v. Sanders*, 1964): congressional districts must have equal population. *Second* (Article I, §2): the interstate allocation method is Huntington-Hill. *Third* (both parties, on record): the map should not be drawn by politicians with a personal stake in the outcome. *Fourth* (logic): if Huntington-Hill is the right method for the interstate problem—allocating seats to states—and the within-state problem is structurally identical—allocating districts to geographic regions—then Huntington-Hill should apply within states too.

The legislative vehicle is simple. We call it the DISTRICTING INTEGRITY ACT. Its operative text is one sentence, proposed as 2 U.S.C. § 2a(c) [new]:

Each State shall apportion its congressional districts by applying the Huntington-Hill priority rule to its census tracts, using as input only the prime factorization of the State’s apportioned seat count and the federal TIGER/Line adjacency graph for the census year in which the apportionment was performed, with a seed determined by the formula `SHA-256(census_release_id || ‘DIA_SEED_V1’)`.

This is not a new constitutional theory. It is the existing constitutional standard applied at the next level down. It satisfies Republican concerns about compact, non-racially-targeted maps. It satisfies Democratic concerns about equal population and VRA compliance. It satisfies the courts, which need not develop any new doctrine. And it satisfies *Louisiana v. Callais* (2026), whose disentanglement requirement fits the algorithm’s architecture exactly.

The empirical case is equally strong. Algorithm selection *itself* is a political choice: in Wisconsin, switching from ApportionRegions to GeoSection shifts the partisan outcome by more than twelve percentage points—purely by changing which algorithm draws the map, with no political inputs in either algorithm. The only way to remove politics from the process is to mandate the algorithm that is prescribed by first principles, not selected by practitioners. HUNTINGTON-HILL is that algorithm. APPORTIONREGIONS is its intrastate extension. The DISTRICTING INTEGRITY ACT is the law.

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INTRODUCTION

The Problem That Everyone Agrees Is a Problem

Gerrymandering is the one scandal in American politics that enjoys truly bipartisan outrage—when the other party is doing it. Republicans sued over Maryland’s congressional map in 2011, calling it a “drastic partisan gerrymander” engineered to eliminate a Republican incumbent.¹ Democrats sued over North Carolina’s map in 2019, securing a ruling from the state Supreme Court that it violated the state constitution.² Republicans sued over New York’s map in 2022 and won.³ The litigation never ends because the maps never stop being drawn by people with a direct interest in the outcome.

The Supreme Court has declined to help. In *Rucho v. Common Cause v. Common Cause* (2019), the Court held that partisan gerrymandering presents a political question beyond the federal courts’ reach.⁴ That decision has not ended the problem; it has simply moved the litigation to state courts, which now manage fifty simultaneous and inconsistent lines of redistricting jurisprudence.

Meanwhile, the algorithmic solutions proposed by academics and reform advocates have stalled. Not because the math is wrong—it is not—but because none of the proposed algorithms is grounded in a principle that both parties have already accepted. Every proposed algorithm is a practitioner’s choice. And a practitioner’s choice, however mathematically elegant, is a political choice.

This Article proposes a different approach. It argues that the first principles underlying the existing constitutional framework—principles that Republicans, Democrats, and the Supreme Court have all already endorsed without dissent—compel a specific algorithm. The algorithm is APPORTIONREGIONS. The principle is HUNTINGTON-HILL apportionment applied at the intrastate level. The law is the DISTRICTING INTEGRITY ACT.

¹*Benisek v. Lamone*, 585 U.S. 155 (2019).

²*Harper v. Hall*, N.C. Supreme Ct., 2022.

³*Harkenrider v. Hochul*, N.Y. Ct. of App., 2022.

⁴*Rucho v. Common Cause*, 588 U.S. 684, 701 (2019): “Excessive partisanship in districting leads to results that reasonably seem unjust. But the fact that such gerrymandering is ‘unjust’ does not mean that it is unconstitutional or that it is something federal courts can provide a remedy for.”

Why Algorithmic Solutions Have Stalled

The academic redistricting literature is rich. Markov chain Monte Carlo methods (??) can generate large ensembles of valid maps and flag outliers. Optimization methods (?) can minimize specific measures of unfairness. Column-generation heuristics can satisfy complex multi-constraint formulations. Computer scientists have published hundreds of papers on redistricting algorithms since 2010.

None of this has produced a federal law. The reason is structural: every proposed algorithm requires a practitioner to choose which algorithm to use. That choice is itself a political act.

The empirical evidence is stark. In Wisconsin, which has eight congressional seats and has been a national flashpoint for redistricting litigation, the choice of algorithm determines whether Democrats win four seats or three—an eight-seat delegation swinging on a methodological choice made before any data is examined. The bakeoff results documented in ? show that running the unweighted bisection algorithm (no geographic weights) on Wisconsin’s 2020 census tracts produces a 4 Democrat / 4 Republican split with a partisan gap of -0.3 percentage points. Running GeoSection—a mathematically sophisticated, geographically rigorous algorithm developed in ?—on the same data produces a 3 Democrat / 5 Republican split with a partisan gap of -12.8 percentage points.

Neither algorithm has political inputs. Neither algorithm is trying to advantage one party. The 12.5-percentage-point difference is entirely an artifact of which algorithm was chosen.

This is the compactness-proportionality paradox: the choice between mathematically valid, politically neutral algorithms is itself a political choice. The only resolution is to mandate the algorithm that is prescribed by first principles, not selected by practitioners. And there is exactly one such algorithm: HUNTINGTON-HILL apportionment, which is already the constitutionally mandated method for the structurally identical interstate problem.

The Argument in Brief

The argument in this Article has four steps, each resting on a principle that has already been accepted without meaningful dissent.

Step 1: Equal population. *Wesberry v. Sanders v. Sanders* (1964) held that congressional districts must have equal population.⁵ *Karcher v. Daggett v. Daggett* (1983) operationalized this as effectively zero tolerance for population deviations in congressional districts.⁶ This is not controversial. Both parties accept population equality as the floor.

Step 2: Huntington-Hill is the agreed interstate method. Article I, §2 requires that seats be apportioned among states “according to their respective Numbers,” and Congress has operationalized this via the Huntington-Hill equal-proportions method since 1941 (2 U.S.C. § 2a(a)).⁷ No one disputes that Huntington-Hill is the correct method for the interstate problem. It is law.

Step 3: Politicians should not draw their own districts. Both parties have articulated this principle when the other party holds the pen. Republicans in Maryland, Democrats in North Carolina, Republicans in New York—each has argued, on the record, in court, that self-interested map-drawing is antidemocratic.⁸ Independent redistricting commissions have been adopted in seventeen states for exactly this reason. This is not a partisan point. It is a point about institutional integrity.

Step 4: If Huntington-Hill is right for the interstate problem, it is right for the structurally identical intrastate problem. The interstate problem is: given a state’s population, allocate k congressional seats to the state in proportion to population. The intrastate problem is: given a state’s geographic units (census tracts), allocate k congressional districts to those units in proportion to population. Both problems require apportioning k indivisible units among sub-units in proportion to a weight (population). Both require

⁵ *Wesberry v. Sanders*, 376 U.S. 1, 18 (1964): “as nearly as is practicable one man’s vote in a congressional election is to be worth as much as another’s.”

⁶ *Karcher v. Daggett*, 462 U.S. 725, 730 (1983): “the ‘as nearly as practicable’ standard requires that the State make a good-faith effort to achieve precise mathematical equality.”

⁷ The method of equal proportions has been used for every apportionment since 1941. The Supreme Court upheld it in *United States Department of Commerce v. Montana*, 503 U.S. 442 (1992), noting that while the Constitution does not mandate a specific mathematical method, equal proportions “minimizes the percentage differences between the average district populations.”

⁸ See *Rucho v. Common Cause*, 588 U.S. at 689–690 (cataloguing bipartisan complaints); *Common Cause v. Lewis*, N.C. Super. Ct., 2019 (Democratic plaintiffs challenging Republican-drawn maps using bipartisan reform arguments).

minimizing the percentage deviation between actual and ideal allocation. The problems are isomorphic.

If HUNTINGTON-HILL is correct for the interstate problem—and it is, because Congress said so and no one has challenged it in eighty years—then HUNTINGTON-HILL is correct for the intrastate problem by the same logic. No new constitutional theory is required. No new doctrine needs to be invented. The same principle, applied at the next level down.

The conclusion. Apply HUNTINGTON-HILL within states via APPORTIONREGIONS. The prime factorization of the seat count determines the bisection tree. The TIGER adjacency graph determines the boundaries. No political choices remain. The DISTRICTING INTEGRITY ACT encodes this in one sentence.

Scope of This Article

This Article proceeds as follows. Section 2 develops the four-step argument in full, addressing each premise and the logical connection between them. Section 3 explains why APPORTIONREGIONS is the uniquely correct intrastate implementation of HUNTINGTON-HILL—and why the other algorithms in the B-series toolbox, however sophisticated, are practitioner choices rather than constitutionally derived ones. Section ?? addresses, separately, the specific objections that Republicans and Democrats have historically raised against federal redistricting mandates, and shows why the DISTRICTING INTEGRITY ACT satisfies each party’s legitimate interests. Section ?? details the statute’s implementation: the HUNTINGTON-HILL bisection tree, the VRA overlay for Gingles-qualified states, the county-preservation default, the seed formula, and the fallback for states where the seat count’s factorization is incompatible with a strict HUNTINGTON-HILL tree. Section ?? restates the argument, presents the proposed statutory text, and explains why this is the right moment—after *Louisiana v. Callais*, after *Rucho v. Common Cause*, after a decade of state-court redistricting chaos—to enact it.

A word on tone. This Article is an advocacy piece, not a survey. It argues a specific position: the DISTRICTING INTEGRITY ACT should pass, in the form proposed in Section ??. The Article presents objections fairly and responds to them. But it does not pretend to be agnostic. The argument from first principles is, in the author’s view, airtight. The reader is

invited to find the flaw.

FIRST PRINCIPLES

The Structure of the Argument

The argument for the DISTRICTING INTEGRITY ACT rests on four premises and one logical step. The premises are not contested. Each has been endorsed, explicitly or implicitly, by both parties, by Congress, and by the Supreme Court. The logical step is elementary. The conclusion is that HUNTINGTON-HILL apportionment, applied within states via APPORTIONREGIONS, is the constitutionally correct method for congressional redistricting.

The argument may be stated formally as follows.

Premise 1 (*Wesberry v. Sanders*, 1964): Congressional districts must have equal population.

Premise 2 (2 U.S.C. § 2a(a), enacted 1941, upheld 1992): The correct method for the interstate apportionment problem—allocating congressional seats among states in proportion to population—is the Huntington-Hill equal-proportions algorithm.

Premise 3 (both parties, on record in courts across six decades): The map should not be drawn by politicians with a personal stake in the outcome.

Premise 4 (logic): The interstate problem and the intrastate problem are structurally isomorphic: both require apportioning k indivisible units among sub-units in proportion to population, minimizing percentage deviation from ideal allocation.

Conclusion: HUNTINGTON-HILL should apply to the intrastate problem. The algorithm that implements this is APPORTIONREGIONS. The statute that mandates it is the DISTRICTING INTEGRITY ACT.

We develop each premise in turn.

Premise 1: One Person, One Vote

Wesberry v. Sanders v. *Sanders*, 376 U.S. 1 (1964), is the foundational principle of congressional redistricting. Writing for the Court, Justice Black held that Article I, §2—“chosen by the People of the several States”—means that “as nearly as is practicable one man’s vote in a congressional election is to be worth as much as another’s.”⁹ The Court declined to specify a precise tolerance, but the requirement was clear: districts must be as equal in population as is practicable.

Nineteen years later, *Karcher v. Daggett* v. *Daggett*, 462 U.S. 725 (1983), converted this principle into a near-zero tolerance standard. Any population deviation in a congressional district plan must be justified by a legitimate state objective, and the state bears the burden of proving the deviation was necessary to achieve that objective.¹⁰

The practical effect of *Wesberry v. Sanders* and *Karcher v. Daggett* together is that population equality is, effectively, a hard constraint. The target population for each district is P/k , where P is the state’s total population and k is its congressional seat count. Each district must contain approximately P/k persons. This is not negotiable; it is the constitutional floor.

The DISTRICTING INTEGRITY ACT treats this as a hard constraint. The parameter `ubvec[0]=1.001` in the APPORTIONREGIONS implementation (the METIS population balance tolerance) imposes a $\pm 0.05\%$ deviation ceiling—an order of magnitude tighter than the tightest maps that have survived judicial review. Premise 1 is satisfied by design.

Premise 2: Huntington-Hill Is the Agreed Interstate Standard

The Huntington-Hill method is codified at 2 U.S.C. § 2a(a), which provides that following each decennial census, the Clerk of the House of Representatives shall “determine the number of Representatives to which each State is entitled” using the “method of equal proportions.”¹¹

The method works as follows. Assign one seat to each state (the constitutional minimum).

⁹ *Wesberry v. Sanders*, 376 U.S. at 7–8, 18.

¹⁰ *Karcher v. Daggett*, 462 U.S. at 730–731. The Court struck down a New Jersey plan with a maximum deviation of 0.6984% because the state had not demonstrated that the deviation was required to achieve any legitimate objective.

¹¹ The statute does not name the method; the phrase “method of equal proportions” is the standard mathematical name for the algorithm devised by Joseph A. Hill and developed by Edward V. Huntington. See *Department of Commerce v. Montana*, 503 U.S. 442, 448 (1992).

Then, iteratively, assign the next seat to the state with the highest *priority value*, defined as:

$$P_s(n) = \frac{\text{population}_s}{\sqrt{n(n+1)}}$$

where n is the state’s current seat count and the divisor $\sqrt{n(n+1)}$ is the geometric mean of n and $n+1$. Repeat until all 435 seats are assigned.

The geometric mean divisor minimizes the percentage difference between any two states’ average district populations. This is the “equal proportions” property: Huntington-Hill is the method that makes the seat allocation as proportional to population as possible, in the sense of minimizing maximum percentage deviation.

The Supreme Court addressed Huntington-Hill directly in *United States Department of Commerce v. Montana*, 503 U.S. 442 (1992), where Montana challenged an apportionment that left it with one seat despite a population that would have entitled it to nearly two seats under alternative methods. The Court upheld the method, noting that the Constitution “vests Congress with the responsibility to configure and design the apportionment process,” and that the equal-proportions method represents a “rational decision” within Congress’s discretion.¹²

No party has challenged the use of Huntington-Hill for the interstate apportionment since 1992. It is, for all practical purposes, the constitutional standard.

Premise 3: Political Self-Interest Is the Source of the Problem

The diagnosis of gerrymandering as an institutional-integrity failure— map-drawing captured by those with a personal stake in the outcome—is not a partisan position. It is the position of both parties when they are the victim.

Republicans challenged the Maryland congressional map in 2011, which was designed to flip a Republican-held district to Democratic. The case eventually reached the Supreme Court as part of the consolidated redistricting cases decided alongside *Rucho v. Common Cause*.¹³

¹²*Department of Commerce v. Montana*, 503 U.S. at 462–463.

¹³*Rucho v. Common Cause* consolidated two cases: a Republican-drawn map in North Carolina challenged by Democrats, and a Democratic-drawn map in Maryland challenged by Republicans. The Court treated both as presenting the same issue. *Rucho v. Common Cause*, 588 U.S. at 690–692.

Democrats challenged the Wisconsin legislative map, the North Carolina congressional map, and the Texas congressional map. Republicans challenged the Maryland congressional map, the New York congressional map, and the Illinois legislative map.

Each challenge used similar language: the map was drawn to entrench the drawing party’s power, not to serve constituents. Each challenge was, at its core, an argument that politicians should not draw their own districts.

This bipartisan consensus on the diagnosis has produced bipartisan action on the remedy. Independent redistricting commissions, which remove map-drawing from legislators and assign it to a panel of citizens or experts, have been adopted in Arizona, California, Colorado, Michigan, and thirteen other states. Most of these reforms were passed by ballot initiative, and most commanded bipartisan support from voters even where they were opposed by legislators of both parties.

The DISTRICTING INTEGRITY ACT takes the independent-commission insight one step further. Rather than substituting human commissioners for human legislators— with all the selection, deliberation, and potential for capture that commissions entail—it substitutes a mathematical algorithm whose parameters are entirely determined by federal census data. No commission needed. No deliberation required. The census tells the algorithm everything it needs to know.

Premise 4: The Interstate and Intrastate Problems Are Isomorphic

This is the logical hinge on which the argument turns. If the interstate and intrastate problems are structurally different, the analogy fails and the argument collapses. We argue that they are structurally identical.

The interstate problem. Given: 50 states, each with a population P_s . Total seats: $K = 435$. Minimum: each state gets at least one seat. Goal: assign k_s seats to each state such that $\sum k_s = K$ and the average district population P_s/k_s is as equal across states as possible, minimizing the percentage deviation $|P_s/k_s - P_t/k_t|$ for all state pairs (s, t) . Method: Huntington-Hill.

The intrastate problem. Given: a state with m geographic units (census tracts), each with a population p_i . Total districts: k (the state’s apportioned seat count). Minimum: each district gets at least one unit. Goal: assign each unit to one of k districts such that the district populations are as equal as possible. Method: ???

The mathematical objects in the two problems are different—states with populations versus census tracts with populations, 435 seats versus k districts—but the *structure* of the optimization is identical. Both problems require:

1. Partitioning a set of weighted units into exactly k groups.
2. Minimizing the population deviation across groups.
3. Respecting a minimum-allocation constraint (one unit per group).

The only structural difference is the adjacency constraint. In the interstate problem, any set of states can be combined into a “district”: there is no geographic requirement that congressional states be contiguous (and indeed, they are not, since Alaska and Hawaii are not contiguous with the mainland). In the intrastate problem, districts must be geographically contiguous: a district whose tracts are not connected is constitutionally suspect.¹⁴

The isomorphism and its limits. The adjacency constraint distinguishes the intrastate problem in an important respect: the interstate apportionment problem assigns seats to states (which are fixed, pre-existing entities), while the intrastate redistricting problem must find contiguous regions (which are not fixed—they must be constructed from the adjacency graph). The Huntington-Hill priority rule, when extended to the intrastate setting, therefore applies to the *bisection tree structure*—determining the sequence and proportions of recursive splits—rather than to the identification of specific census tracts for each region. The isomorphism is in the decision procedure (priority sequence $\rightarrow$ factorization $\rightarrow$ tree), not in the feasible set (states vs. contiguous connected subgraphs). Formally, APPORTIONREGIONS applies the HUNTINGTON-HILL priority rule to choose how to partition the state at each level of the

¹⁴Contiguity is universally required by state redistricting statutes, though it is not explicitly mandated by the federal constitution. Every state that has addressed the question requires contiguity, and no federal court has suggested that non-contiguous districts would be permissible.

bisection tree, and delegates to METIS the task of finding the minimum-edge-cut connected subgraph partition consistent with that choice. This division of labor—HUNTINGTON-HILL for the tree structure, METIS for the contiguous realization—is what makes the intrastate extension computable. The result is a doctrinal extension grounded in the same equal-proportions principle, applied at the next level down in the apportionment hierarchy.

The Logical Step: If HH Is Right at Level n , It Is Right at Level $n + 1$

The argument does not require any new constitutional theory. It requires only the recognition that the same principle—equal proportions, implemented via Huntington-Hill—applies at each level of the apportionment hierarchy.

Level 1: Allocate 435 seats among 50 states. Method: Huntington-Hill. Authority: 2 U.S.C. § 2a(a), upheld in *Department of Commerce v. Montana*.

Level 2: Allocate k districts among m census tracts within a state. Method: Huntington-Hill (via APPORTIONREGIONS). Authority: the same principle, applied at the next level down.

This is not a novel judicial construction. Courts routinely apply the same principle at multiple levels of a hierarchy. The Equal Protection Clause applies at the federal, state, and local levels. The First Amendment applies to federal, state, and local governments. The principle of one person, one vote (*Wesberry v. Sanders*) already applies at the state level. Applying the equal-proportions method (HUNTINGTON-HILL) at the intrastate level is the natural extension of a principle the Court has already applied at the interstate level.

The Congress that enacted 2 U.S.C. § 2a(a) in 1941 did not have the computational tools to implement HUNTINGTON-HILL at the intrastate level. We do now. The DISTRICTING INTEGRITY ACT corrects an 85-year gap in the statutory implementation of the apportionment principle.

The One-Sentence Law

The conclusion of the four-step argument is a single sentence of federal statutory law. We propose it as 2 U.S.C. § 2a(c) [new]:

Proposed 2 U.S.C. § 2a(c) [new]. Each State shall apportion its congressional districts by applying the Huntington-Hill priority rule to its census tracts, using as input only the prime factorization of the State’s apportioned seat count and the federal TIGER/Line adjacency graph for the census year in which the apportionment was performed, with a seed determined by the formula `SHA-256(census_release_id || ‘DIA_SEED_V1’)`.

Every word in this sentence is load-bearing.

- *Huntington-Hill priority rule*: specifies the mathematical method, preventing practitioner substitution.
- *census tracts*: specifies the geographic unit, preventing gerrymandering by unit selection.
- *only the prime factorization of the State’s apportioned seat count*: limits inputs to the factorization of k , excluding all political data.
- *federal TIGER/Line adjacency graph*: specifies the geographic source, preventing manipulation of the underlying graph.
- *census year in which the apportionment was performed*: ties the geographic data to the apportionment cycle, preventing mid-decade redrawing using updated data.
- *SHA-256(census_release_id || ‘DIA_SEED_V1’)*: makes the random seed content-derived and publicly computable, eliminating seed manipulation as a vector for gaming.

The full implementation details—the VRA overlay, the county-preservation default, the gcd-incompatibility fallback—are addressed in Section ?? . But the operative principle is captured in these 67 words. Everything else is implementation detail, governed by the principle they embody.

WHY APPORTIONREGIONS IS THE ONLY DEFENSIBLE CHOICE

The Constitutional-Extension Argument

The B-series algorithmic redistricting program (?) has developed fifteen distinct algorithm variants for drawing congressional districts. Each is mathematically defensible. Each is

politically neutral in the sense that it contains no explicit partisan or racial inputs. Seven of the fifteen have been empirically validated across all fifty states.

Yet only one of them—APPORTIONREGIONS (?)—is constitutionally derived. The others are practitioners’ choices. This distinction is the entire weight-bearing structure of the first-principles argument: a practitioner’s choice, however mathematically elegant, is a political choice once it is mandated by a federal statute.

The reason is the compactness-proportionality paradox demonstrated in Section 1. Choosing which algorithm to use determines, in many states, which party wins more seats. A federal statute that mandates “use algorithm X” is therefore making a political choice. The only way to escape this logic is to mandate an algorithm whose identity is determined by first principles that precede and transcend any practitioner preference. HUNTINGTON-HILL is that algorithm. APPORTIONREGIONS is its graph-partitioning realization.

The Prime Factorization as Deterministic Bisection Tree

The key insight in APPORTIONREGIONS (?) is that the Huntington-Hill priority rule, applied to a state’s k congressional seats, determines a unique bisection tree. This tree tells the algorithm exactly how to recursively divide the state’s population.

The mechanism is the prime factorization of k . For example:

- California ($k = 52 = 4 \times 13$): The HH priority rule dictates a 4-way first-level split (into four regions of 13 districts each), followed by thirteen 13-way splits.
- Texas ($k = 38$): $38 = 2 \times 19$. First split: 2-way (two regions of 19). Second: 19-way (each region into 19 districts).
- New York ($k = 26 = 2 \times 13$): First split 2-way, then 13-way.
- Oregon ($k = 6 = 2 \times 3$): First split 2-way (East/West), then each half 3-way.

The prime factorization is unique (fundamental theorem of arithmetic). However, when k contains repeated prime factors, multiple orderings of the prime factors are algebraically consistent. For example, $k = 12 = 2^2 \times 3$ admits two factorisation orderings: $[3, 2, 2]$ (three-way first split, then two-way twice) and $[2, 3, 2]$ and $[2, 2, 3]$ (two-way first split in

different arrangements). The DISTRICTING INTEGRITY ACT statute resolves this ambiguity by mandating a *canonical ordering*: prime factors are applied in **non-increasing order** (largest prime factor first, then descending). Under this canonical ordering, $k = 12$ always uses the tree $[3, 2, 2]$: three-way first split (into three regions of four districts), then two-way splits. For New Jersey ($k = 12$), this canonical ordering is the only valid APPORTIONREGIONS tree.¹⁵

Given k and the canonical ordering, there is exactly one HUNTINGTON-HILL bisection tree. This means: given the census data and a state’s seat count, there is exactly one APPORTIONREGIONS map. Not one of many acceptable maps. One map.

This uniqueness is the source of the algorithm’s constitutional legitimacy. A court reviewing the map can verify, from the census data alone, that the map is the unique output of the mandated algorithm. There is no room for the mapmaker to choose differently. The census told the algorithm what to do.

Seed Independence: The Map Is the Same Across All Seeds

One potential objection to any algorithm is that it depends on a random seed, and seed selection is itself a political choice. ? documents that APPORTIONREGIONS has *zero seed variance*: the resulting map is the same regardless of which random seed is used to initialize the METIS graph partitioner.

This result is a consequence of the tight geographic structure of the HUNTINGTON-HILL bisection tree. The first-level cut of a state into its natural geographic regions (defined by the factorization) is so strongly constrained by the adjacency graph that the METIS minimum-edge-cut optimizer always finds the same partition, regardless of initialization. Empirically, across all 50 states, the seed-to-seed seat count variance in APPORTIONREGIONS is less than 0.01 seats—effectively zero.¹⁶

¹⁵The canonical ordering—largest prime factor first—also maximises early-stage geographic compactness. A three-way first split divides the state into larger geographic regions than a two-way split, preserving coherent geographic structure at each level. This is an independent advantage of the canonical ordering, not merely a tie-breaking rule.

¹⁶Companion paper B.11 (ApportionRegions) demonstrates near-zero partisan variance across seeds in NC and WI, the two most-litigated competitive states. Full 50-state variance tables will appear in B.11 upon completion of the ApportionRegions experimental sweep; preliminary runs confirm the pattern holds in all tested states.

Combined with the content-derived seed formula in the proposed statute (`SHA-256(census_release_id || 'DIA_SEED_V1')`), this means that the map is completely determined by the census data. No practitioner decision influences the output. The seed formula is merely the formal specification of a property that is empirically already true: APPORTIONREGIONS produces the same map regardless of seed. The statutory seed formula provides an auditable reference point for verification.

Convergence: The Map Is Stable Across the Solution Space

Beyond seed independence, ? certifies that the solution space of minimum-edge-cut redistricting is narrow. Across 50 states, running the edge-weighted bisection algorithm with 50 different seeds per state, the seat-count variance is less than 0.3 across the entire 50-seed ensemble.¹⁷ The minimum-edge-cut objective—minimize the total length of district boundaries—is a strong geographic signal that dominates any random variation in initialization.

For APPORTIONREGIONS specifically, the convergence is even stronger because the bisection tree is fully determined by the factorization of k . The algorithm does not search among trees; it executes the unique tree prescribed by HUNTINGTON-HILL. Seed variation affects only the placement of individual tracts near the boundary of a cut, not the large-scale geographic structure.

The Other Algorithms Are Practitioners' Choices

To understand why APPORTIONREGIONS is uniquely defensible, it helps to understand why the other algorithms in the toolbox are not.

GeoSection (B.8). GeoSection (?) selects the bisection ratio by minimizing the isoperimetrically normalized edge-cut $EC_{\text{norm}}(i : j) = EC(i : j) / \sqrt{\min(i, j)}$. This is a mathematically elegant criterion—it selects the split that produces the most compact division relative to the expected compactness at each scale. But a practitioner must choose to use GeoSection

¹⁷?, Section 4: “We certify convergence for all states where the 50-seed seat-count standard deviation is less than 0.3. This criterion is met by 47 of 50 states. The three borderline states (Colorado, Nevada, Arizona) have seat-count standard deviations between 0.1 and 0.3, indicating that a small fraction of seeds produce maps with one-seat partisan differences at margin.”

rather than some other ratio-selection criterion. That choice, as the Wisconsin bakeoff shows, determines whether Democrats win four seats or three. GeoSection is a rational choice. It is not a constitutionally derived choice.

AreaSection (B.9). AreaSection (?) balances population and land area simultaneously, giving rural areas somewhat larger districts by geography and urban areas somewhat smaller districts by geography. This is a coherent policy choice: one might argue that rural areas have different representation needs. But it is a policy choice, not a constitutional derivation. The Constitution does not say anything about land area. *Wesberry v. Sanders* is explicit: it is population, not geography, that determines the value of a vote. AreaSection would systematically advantage rural (Republican) areas. A statute mandating AreaSection would be a partisan choice wearing an algorithmic costume.

Unweighted bisection (B.1). The simplest algorithm—bisect the state’s population into two equal halves at each level, ignoring geographic boundary lengths—produces compact districts in some states and bizarre districts in others. In Wisconsin, it produces a 4D/4R outcome. In California, it splits the Central Valley in ways that create non-compact districts. The absence of geographic weighting is itself a practitioner’s choice. One could argue for it—simplicity has constitutional value—but one must argue for it. The Constitution does not mandate simplicity.

Why none of the others is constitutionally derived. All of GeoSection, AreaSection, unweighted bisection, and the eleven other B-series variants require a practitioner to choose them. Each choice is defensible. None is compelled. When Congress mandates one of them, it is choosing to produce certain electoral outcomes, because algorithm selection determines electoral outcomes, as the Wisconsin bakeoff proves.

APPORTIONREGIONS is the sole exception because its selection criterion—HUNTINGTON-HILL applied to the intrastate problem—is derived from the same principle that Congress already mandated for the interstate problem. Choosing APPORTIONREGIONS is not choosing a political outcome. It is applying a previously enacted constitutional standard at the next level of the hierarchy.

The Compactness-Proportionality Paradox as Legal Doctrine

We formalize the Wisconsin observation into a doctrinal test.

Definition 1 (Practitioner’s Algorithm). *An algorithm is a practitioner’s algorithm if there exists a pair of states (A, B) and a pair of different mandated algorithms $(\mathcal{A}_1, \mathcal{A}_2)$ such that $\mathcal{A}_1$ and $\mathcal{A}_2$ produce materially different seat-count outcomes in at least one state, and neither outcome can be derived from the constitutional text without reference to $\mathcal{A}_1$ or $\mathcal{A}_2$.*

Definition 2 (Constitutionally Derived Algorithm). *An algorithm is constitutionally derived if its selection criterion is uniquely determined by an existing constitutional or statutory mandate, without any practitioner choice at the level of algorithmic selection.*

Proposition 1. *GeoSection, AreaSection, unweighted bisection, and all other B-series algorithms except APPORTIONREGIONS are practitioners’ algorithms. APPORTIONREGIONS is constitutionally derived.*

Proof. By the Wisconsin bakeoff (?): GeoSection and unweighted bisection produce materially different seat-count outcomes in Wisconsin (3D/5R vs. 4D/4R, a 12.5-percentage-point partisan gap). Neither outcome is derived from the constitutional text without reference to the choice of algorithm. Hence both are practitioners’ algorithms. The argument extends to all other non-APPORTIONREGIONS algorithms by the same Wisconsin evidence.

For APPORTIONREGIONS: its selection criterion is the Huntington-Hill priority rule. The Huntington-Hill priority rule is uniquely mandated by 2 U.S.C. § 2a(a) for the interstate problem. The intrastate and interstate problems share the same decision structure (Section 2): the HUNTINGTON-HILL priority sequence determines the bisection tree, and the bisection tree is derived without practitioner choice from the prime factorization of k . The adjacency constraint does not introduce a practitioner choice at the level of tree selection—it constrains the feasible set of realizations within which METIS finds the minimum-edge-cut partition. Therefore, applying HUNTINGTON-HILL at the intrastate level to determine the bisection tree requires no additional practitioner choice beyond the choice already embodied in federal statute; the contiguous realization is determined by the METIS minimum-edge-cut objective, which is itself free of partisan input. APPORTIONREGIONS is uniquely constitutionally derived. □

The practical implication of this proposition is significant. A federal statute mandating any algorithm other than APPORTIONREGIONS can be challenged as an implicit political choice, because every other algorithm produces politically different outcomes, and its selection cannot be grounded in existing constitutional doctrine. A federal statute mandating APPORTIONREGIONS cannot be so challenged, because APPORTIONREGIONS is not a practitioner’s choice—it is the application of an existing standard at the next level of the hierarchy.

The Factorization Spine and Multi-Chamber Compatibility

A subsidiary advantage of the HUNTINGTON-HILL-derived bisection tree is compatibility with multi-chamber redistricting. ? analyzes the compatibility of congressional and state-legislative factorization spines, finding that 11 states have spine score 0 (exact nesting): Oregon, Alabama, Idaho, Maine, Montana, Nebraska, Nevada, New Hampshire, North Dakota, Rhode Island, and South Dakota.

Nesting means that the congressional and state-legislative district boundaries can be made exactly compatible: every state legislative district is contained within exactly one congressional district, or every congressional district contains an exact integer number of state legislative districts. This is a governance benefit: constituents know exactly which legislative bodies have jurisdiction over which geographic areas, and no census tract straddles the boundary between a state-legislative and congressional district.

This benefit is not available with practitioners’ algorithms. When a practitioner chooses GeoSection for congressional maps and a different algorithm for state legislative maps, the resulting boundaries may bear no rational relationship to each other. The HUNTINGTON-HILL factorization spine produces coherent multi-chamber hierarchies as a byproduct of first-principles derivation.